$P = 0.126513 6N$

xxx

Wedge: Members used to lift masses.

$$\eta = \frac{W}{P}$$

$P \downarrow \quad \eta \uparrow$

11/04/23

⇒ **Multistart thread:**

distance { Pitch: ~~alternate~~ side by side
Lead: same colour (i.e. in one rotation)

(Screw jack)

⇒ **V-belt drive**

⇒ **Cross belt drive**

⇒ **Open-belt drive**

⇒ **Flat belt drive**

Thread cut on a cylindrical surface is called screw.

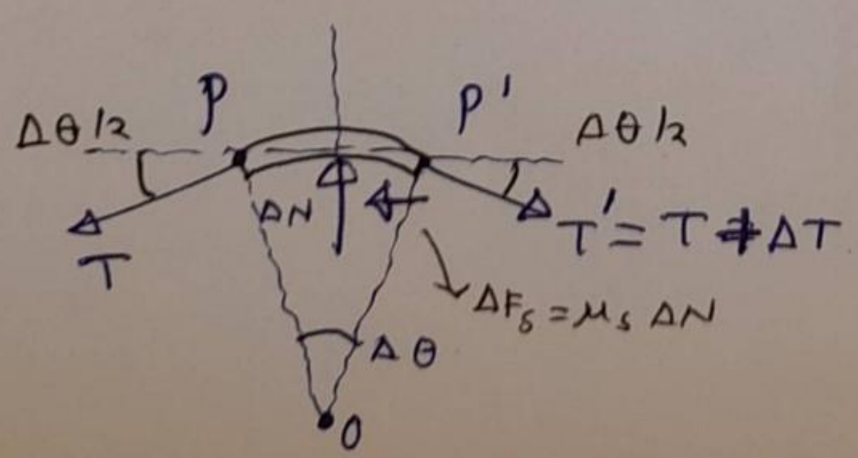
Lead = n × Pitch

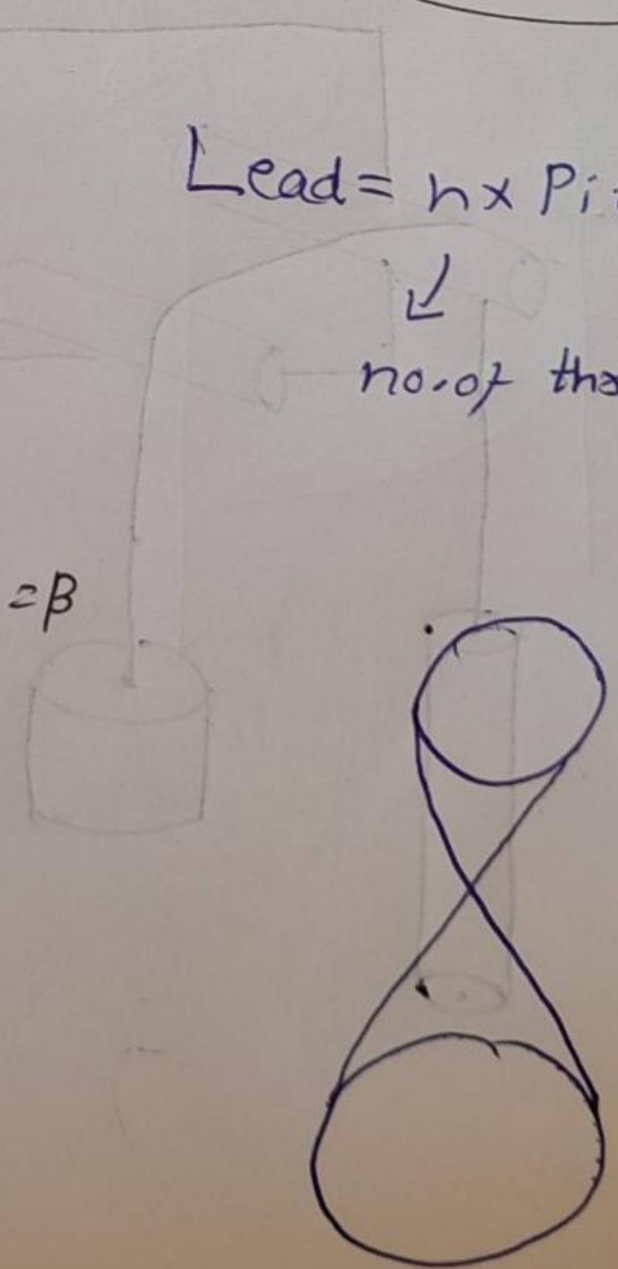
n → no. of threads

Belt Friction

Friction $\propto \beta$

Angle of friction $= \beta$





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$$\Sigma F_x = 0: \ (T+\Delta T)\cos\frac{\Delta\theta}{2} - T\cos\frac{\Delta\theta}{2} - \mu_s \Delta N = 0$$

$$\Sigma F_y = 0: \ \Delta N - (T+\Delta T)\sin\frac{\Delta\theta}{2} - T\sin\frac{\Delta\theta}{2} = 0$$

$$\frac{\Delta T}{\Delta\theta}\underbrace{\cos\frac{\Delta\theta}{2}}_{\to 1} - \mu_s\left(T + \frac{\Delta T}{2}\right)\underbrace{\frac{\sin(\Delta\theta/2)}{\Delta\theta/2}}_{\lim_{\Delta\theta\to 0} \to 1}$$

$$\boxed{\frac{dT}{d\theta} - \mu_s T = 0}$$

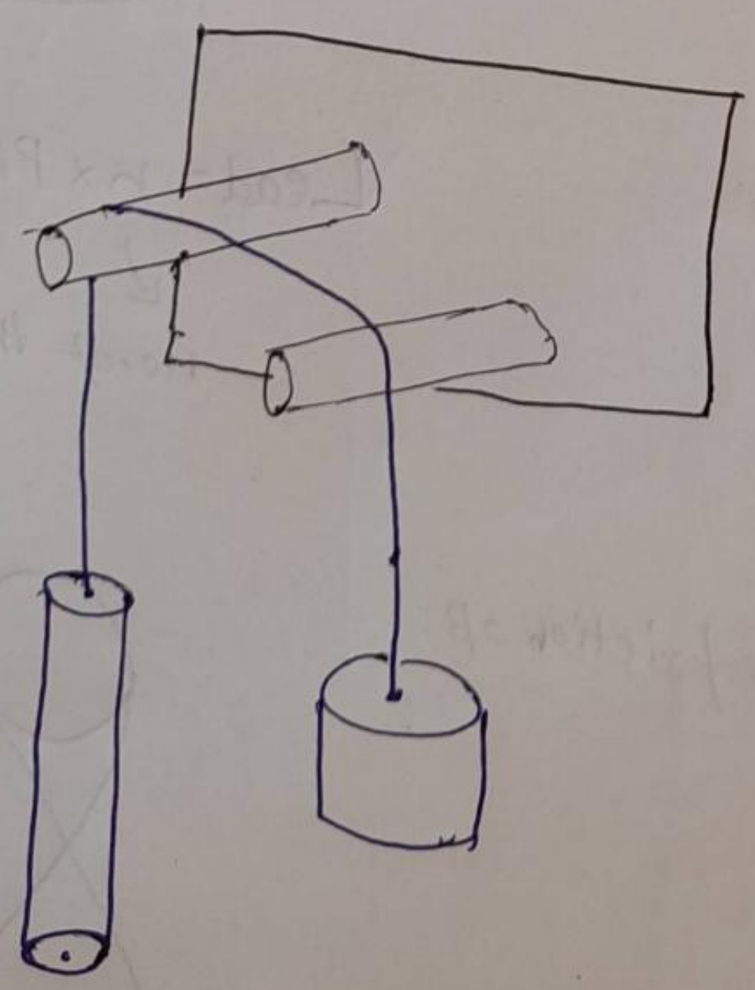
$$\int_{T_1}^{T_2} \frac{dT}{T} = \int_{\theta=0}^{\theta=\beta} \mu_s \, d\theta$$

$$\ln\frac{T_2}{T_1} = \mu_s \beta \Rightarrow \boxed{T_2 = T_1 e^{\mu_s \beta}}$$

$$\Rightarrow \frac{T_2}{T_1} = e^{\mu_s B}$$

Power delivered $= (T_2 - T_1)\omega = T_1(e^{\mu_s \beta} - 1)\omega$ //

Q.11



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